二、旋转矩阵

2.1 旋转矩阵的特性

2.1.1 旋转矩阵的正交性

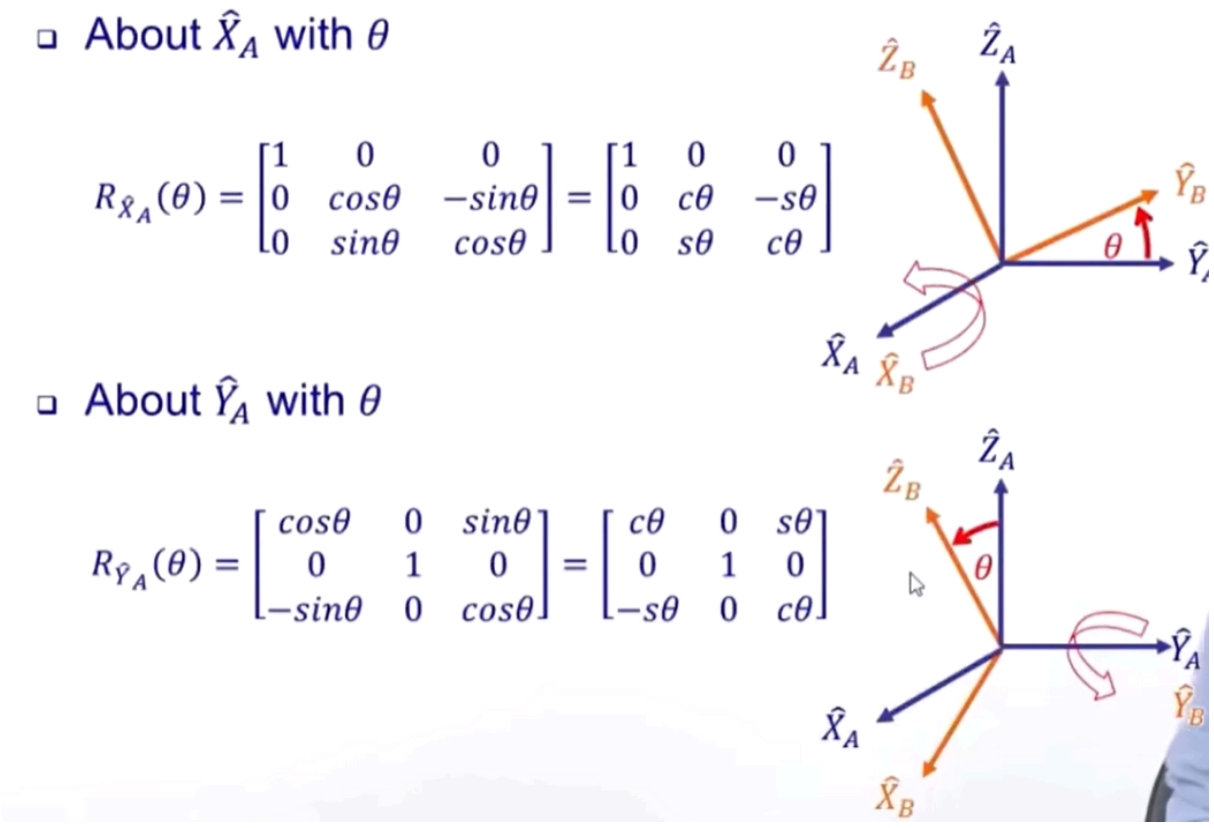
- 推导：从 {A} 到 {B} 的旋转矩阵 ${}^{A}_{B}R$ 等于${}^{B}_{A}R$的转置。
- 核心结论：

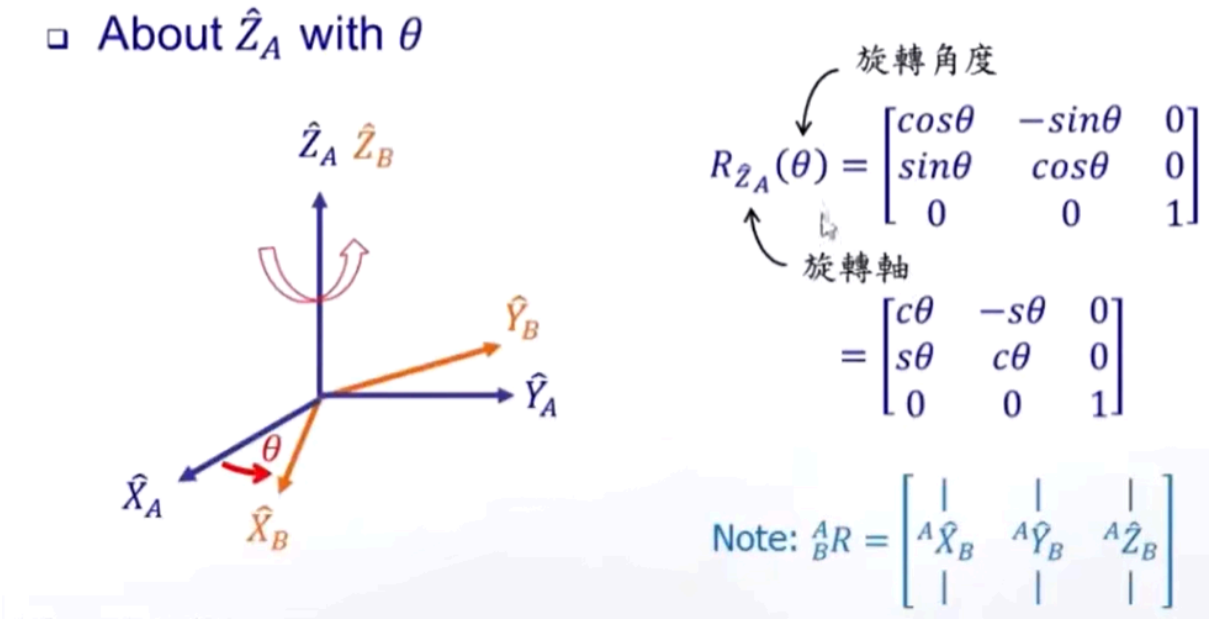
$$ {}^{A}_{B}R = {}^{B}_{A}R^{T} $$

即旋转矩阵是**正交矩阵**，其逆矩阵等于其转置矩阵。

- 同时：

$$ {}^{A}_{B}R^{T} = {}^{A}_{B}R^{-1} $$

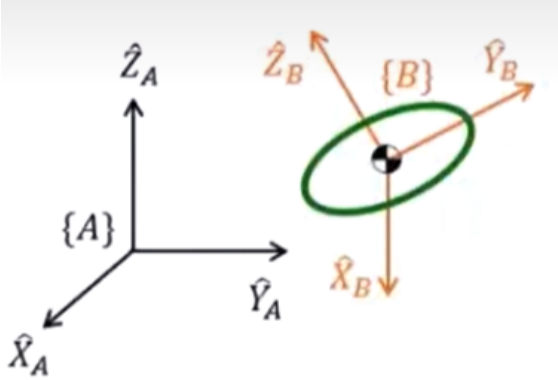




2.2 旋转矩阵的三种用法

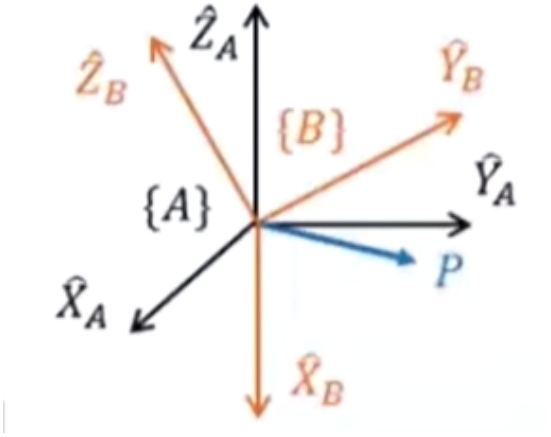
1.描述一个 frame（相对于另一个 frame）的姿态

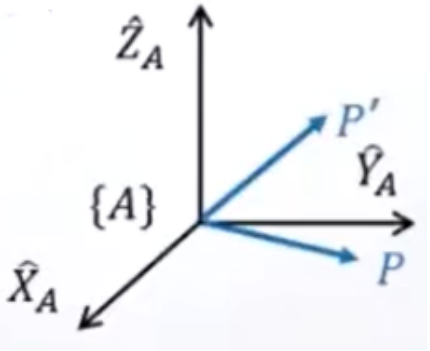
$$ {}^A_BR = \begin{bmatrix} | & | & | \\ {}^A\hat{X}_B & {}^A\hat{Y}_B & {}^A\hat{Z}_B \\ | & | & | \end{bmatrix} $$



2.将 point 由某一个 frame 的表达转换到另一个和此 frame 仅有相对转动的 frame 来表达

$$^{A}P = {}^{A}_{B}R\,{}^{B}P$$



3.point（vector）在同一个 frame 中进行转动

$$^{A}P' = R(\theta)\,{}^{A}P$$